

Solution 4

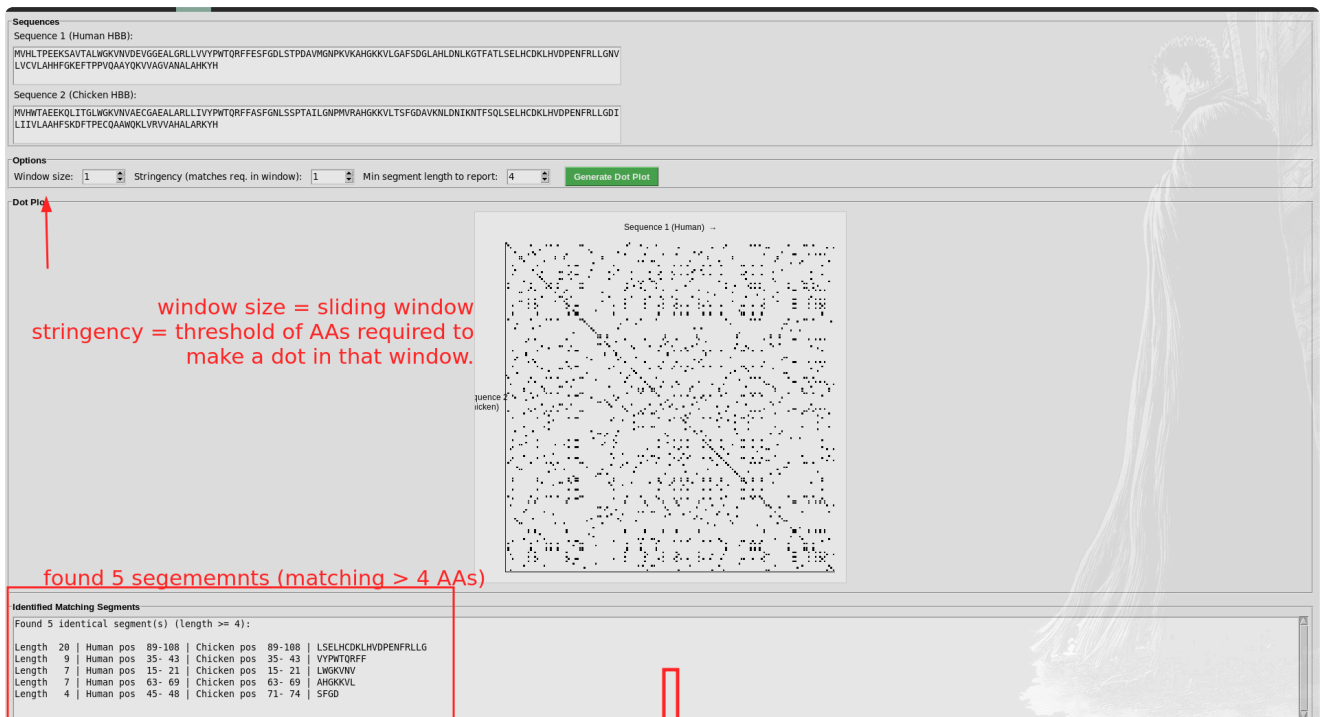
BS24B006

Apoorv Shah

all codes given in github [link](#)

Q1

code



at window size 5 and threshold 3, its much cleaner

Sequences

Sequence 1 (Human HBB):

MVHLTPEKSAVTALWGKVVDEVGGEALGRLLVYYPWTQRFESFGDLSTPDVWGNPKVKAHGKKVLGAFSDGLAHLDNAIKGTATLSELHCDKLHVDPENFRLLGQV
LVCLVAHFGKEFTTPVQAAVKVAVGAVNALAHKYH

Sequence 2 (Chicken HBB):

MVHNTAEKQLITGLWGKVVVAEGGAELARLLTVYPWTQRFASFQNLSSPTAILGNPMVRAHGKKVLTSGDAVKNLDNAIKNTFSQSELHCDKLHVDPENFRLLGDI
LTIIVLAHFSGDFTPECQAAMQKLVRRVVAHALARKYH

Options

Window size: 5

Stringency (matches req. in window): 3

Min segment length to report: 4

Generate Dot Plot

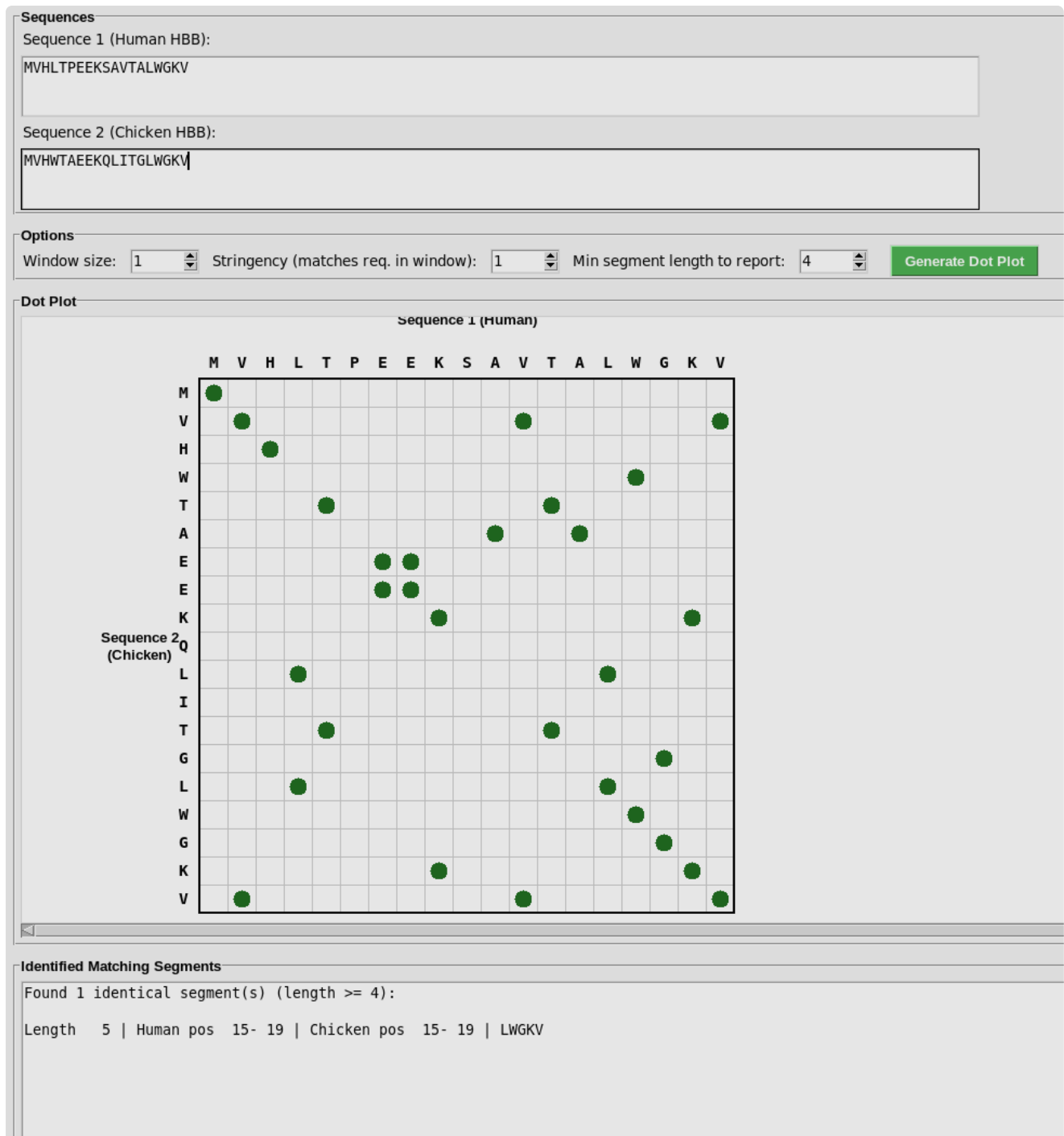
Dot Plot

Identified Matching Segments

Found 5 identical segment(s) (length >= 4):

Length 20	Human pos 89-108	Chicken pos 89-108	LSELHCDKLHVDPENFRLLG
Length 9	Human pos 35-43	Chicken pos 35-43	VYPWTQRF
Length 7	Human pos 15-21	Chicken pos 15-21	LWGKVV
Length 7	Human pos 63-69	Chicken pos 63-69	AHGKKVL
Length 4	Human pos 45-48	Chicken pos 71-74	SFGD

we can see the first 20 residues as well



(for small snippets of AA it gives a detailed more visible plot)

Q2

assignment_4 >  q2_score.py > ...

```
1  seq1 = "AATCTATA"
2  seq2 = "AAG--ATA"
3
4  match_score = 1
5  mismatch_score = 0
6  origination_penalty = -2
7  length_penalty = -1
8
9  score = 0
10 in_gap = False
11
12 for a, b in zip(seq1, seq2):
13     if a == '-' or b == '-':
14         if not in_gap:
15             score += origination_penalty
16             in_gap = True
17         score += length_penalty
18     else:
19         in_gap = False
20         if a == b:
21             score += match_score
22         else:
23             score += mismatch_score
24
25 print("Total alignment score:", score)
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

 Python + v   ... |

```
    ---(_)
   (/  ---\
  (/ |   |
   \   ---/
    ---(_)
```

OS: Ubuntu 22.04.5 LTS x86_64
Kernel: 6.8.0-138-generic
Uptime: 6 days, 6 hours, 7 mins
Shell: zsh 5.8.1

• > /usr/bin/python3 /home/specapoorv/bioinfo/assignment_4/q2_score.py

Total alignment score: 1

○ ~/bioinfo main ?5 > |

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q.3

doing manually now

Col	seq1	seq2	Type	Score
1	A	A	match	+1
2	A	A	match	+1
3	T	G	mismatch	0
4	C	-	gap start	-2 (origination) -1 (length) = -3
5	T	-	gap continue	-1
6	A	A	match	+1
7	T	T	match	+1
8	A	A	match	+1

sum = 1 + 1 + 0 - 3 - 1 + 1 + 1 + 1 = 1 (same so code works)

q.4

all codes given in github [link](#)

q4 code

```

> /usr/bin/python3 /home/specapoorv/bioinfo/assignment_4/q4_needleman.py
Optimal Alignment:
ACAGTCGAACG
ACCGTC---CG
Optimal Alignment Score: 7
~/bioinfo main ?5 >
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```

q.5

		A	C	C	G	T	C	C	G
	0	-2	-4	-6	-8	-10	-12	-14	-16
A	-2	12	0	-2	-4	-6	-8	-10	-14
C	-4	0	14	12	0	-2	-4	-6	-8
A	-6	-2	2	13	1	-1	-3	-5	-7
G	-8	-4	0	1	15	3	1	-1	-3
T	-10	-6	-2	-1	3	7	5	3	1
C	-12	-8	-4	0	1	5	9	7	5
G	-14	-10	-6	-2	2	3	7	18	9
A	-16	-12	-8	-4	0	1	5	6	7
A	-18	-14	-10	-6	-2	-1	3	4	5
C	-20	-16	-12	-8	-4	3	1	5	3
G	-22	-18	-14	-10	-6	-5	-1	3	7

q.6

code

```

> /usr/bin/python3 /home/specapoorv/bioinfo/assignment_4/q6_smith.py
Best Local Alignment:
CGTATCG
CGTATCG
Score: 14
Full alignment view:
Seq1: ACGTATCGCGTATA
Seq2: -CGTATCG-----
~/bioinfo main !1 ?2 >

```

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